Tracing Policy Priorities in Nepal: A Comparative Topic Modeling Analysis of Government Five-Year Plans and Budget Speeches

Sandesh Tamang
Softwarica College of IT and E-Commerce
in collaboration with Coventry University
Kathmandu, Nepal

Aishwarya Rai
Softwarica College of IT and E-Commerce
in collaboration with Coventry University
Kathmandu, Nepal

Abstract—We examine seventeen years of Nepali development discourse by combining Latent Dirichlet Allocation (LDA) with a Gaussian Process (GP) classifier and regressor. A corpus of three Periodic Plans published by the National Planning Commission (2007–2024) and eight annual budget speeches delivered by the Ministry of Finance is split into 345 fixed-length chunks and modelled with eight latent topics. The discovered themes – fiscal management, macroeconomic policy, trade and industry, infrastructure, human development, governance, agriculture, and federal allocation – align closely with the substantive content of Nepal's planning literature. Using the per-chunk topic distributions as features, a GP classifier with an RBF kernel separates Plans from Budget Speeches at 90.7% ± 2.8% five-fold cross-validated accuracy, exceeding a logistic-regression baseline by roughly nine accuracy points. A GP regressor with a Matérn-2.5 kernel further predicts the year of authorship to within 3.48 years on average ($R^2 = 0.353$), exposing measurable thematic drift over the federal-transition period. The study illustrates how unsupervised text mining and probabilistic supervised models can be combined to quantify policy continuity and change in a low-resource governance setting, and it releases a reproducible pipeline for follow-up work.

Index Terms—Topic modelling, Latent Dirichlet Allocation, Gaussian Process, text mining, public policy, Nepal, budget speech, periodic plan.

I. Introduction

Periodic Plans and annual budget speeches are the two flagship texts through which the Government of Nepal communicates its development intent. Periodic Plans, published every three to five years by the National Planning Commission, set medium-term sectoral targets, while budget speeches operationalise those targets through fiscal-year allocations announced by the Minister of Finance. Together they form a long, dense and surprisingly under-analysed record of the country's policy priorities – a record that spans monarchy, civil conflict, constitutional change and the on-going federal transition.

Most existing scholarship on Nepal's plans is qualitative and relies on close-reading of individual documents. Quantitative comparative work is rare, partly because the documents are long (the Fifteenth Plan alone exceeds 180,000 words) and partly because they are scattered across several ministry portals in heterogeneous formats. In this paper we treat the corpus as a single textual data set and ask three questions:

1) Which latent themes recur across plans and speeches, and in what proportions?
2) Can a probabilistic supervised model decide, from theme composition alone, whether an unseen passage came from a Plan or from a Budget Speech?
3) Do the thematic mixtures themselves carry a temporal signature strong enough to date a passage?

We answer the first question with Latent Dirichlet Allocation (LDA) [1], the second with a Gaussian Process (GP) classifier [2] on the per-document topic distributions, and the third with a GP regressor whose kernel encodes smoothness assumptions appropriate for a slowly evolving discourse. Section II situates the work in the topic-modelling and GP literature, Section III describes the corpus and its provenance, Section IV formalises the methods, Section V reports the experimental setup, Section VI presents the empirical findings, Section VII discusses social and ethical considerations, and Section VIII concludes.

II. Related Work

Topic modelling has been widely applied to legislative and budgetary text. Quinn et al. [3] use a dynamic topic model to track U.S. Senate attention over fifty years; Grimmer and Stewart [4] survey the broader field; Lucas et al. [5] extend the methodology to multilingual political corpora. In the development-policy domain, work on World Bank project narratives [6] and on Indian five-year plans has shown that latent themes can be tracked through time even when the underlying document genre is highly formulaic.

For Nepal specifically, the literature is sparse. Existing quantitative work has focused either on macroeconomic outcomes implied by the plans or on hand-coded sectoral content analyses. To our knowledge no published study has combined LDA with a Gaussian Process discriminative layer on the Nepali periodic-plan/budget-speech corpus.

Gaussian Process classification and regression remain attractive in this regime because (i) the input dimensionality after LDA is small (eight topics), (ii) the number

TABLE I: Source documents in the corpus.

Document	Year	Words
11th Three-Year Interim Plan	2007	177,025
13th Plan, Approach Paper	2013	58,296
15th Periodic Plan	2019	186,314
Budget Speech FY2008/09	2008	40,195
Budget Speech FY2014/15	2014	36,718
Budget Speech FY2016/17	2016	43,302
Budget Speech FY2017/18	2017	38,626
Budget Speech FY2018/19	2018	24,632
Budget Speech FY2021/22	2021	38,390
Budget Speech FY2023/24	2023	24,005
Budget Speech FY2024/25	2024	17,441
Total		684,944

of training units is modest (a few hundred chunks), and (iii) the posterior gives calibrated uncertainty in addition to a point prediction [2], [7]. The standard squared-exponential (RBF) and Matérn kernels provide contrasting smoothness priors that are particularly useful when, as here, the analyst has weak prior beliefs about the geometry of the feature space [8].

III. Problem and Data Set

A. Documents collected

Three Periodic Plans (Eleventh Three-Year Interim Plan 2007/08–2009/10, Thirteenth Plan 2013/14–2015/16, and Fifteenth Plan 2019/20–2023/24) were downloaded in their official English versions from the National Planning Commission and from public mirrors maintained by ICIMOD, the Asian Development Bank, and FAOLEX. Eight budget speeches covering fiscal years 2008/09 through 2024/25 were downloaded from the Ministry of Finance archive (mof.gov.np) and verified against alternative mirrors when the primary link returned an HTTP error. Table I summarises the collected material.

B. Why these documents matter

The chosen window straddles three regimes that should, in principle, leave detectable thematic fingerprints: the post-conflict transitional government (2007–2014), the promulgation of the 2015 Constitution and the move to federalism (2015–2017), and the consolidation phase under federal Province governments (2018–2024). An empirical method that can or cannot pick these regimes apart from the documents' internal language is therefore informative regardless of outcome.

C. Pre-processing

The documents are PDFs, so we use pypdf to extract raw character streams, re-join hyphenated line breaks, collapse repeated whitespace, and cast everything to lower case. Each cleaned document is partitioned into non-overlapping chunks of $\sim$2,000 words to expose enough units (345 in total) for stable variational inference. Tokens are extracted with the regular expression [a-z]{3,}, the NLTK English stopword list is applied, and a manually-curated list of 70 boilerplate terms specific to Nepali government English (nepal, government, ministry, fiscal, rupees, ...) is removed so that they do not dominate the topics. The resulting vocabulary has 4,000 terms after enforcing min = 3 document frequency and max = 0.85 document frequency.

IV. Methods

A. Latent Dirichlet Allocation

LDA models each document d as a mixture over K latent topics $\theta_d \sim \mathrm{Dir}(\alpha)$ and each topic k as a distribution over the vocabulary $\phi_k \sim \mathrm{Dir}(\beta)$. A token w_{dn} is generated by first drawing a topic $z_{dn} \sim \mathrm{Cat}(\theta_d)$ and then a word $w_{dn} \sim \mathrm{Cat}(\phi_{z_{dn}})$. We fit LDA with batch variational Bayes as implemented in scikit-learn, using $\alpha = 0.1$ (favouring sparse document-topic mixtures) and $\beta = 0.01$ (favouring sparse topic-word distributions), 40 outer iterations and a fixed random seed.

B. Gaussian Process classifier

After LDA we obtain a feature matrix $\Theta \in \mathbb{R}^{N\times K}$, where $\Theta_{d,k}$ is the expected proportion of topic k in chunk d. We model the binary label $y_d \in \{\mathrm{plan}, \mathrm{speech}\}$ with a GP classifier whose latent function f has prior $f \sim \mathrm{GP}(0, k(\cdot,\cdot))$ and likelihood $p(y \mid f) = \Phi(yf)$. The kernel used is

$$k(x, x') \;=\; \sigma_f^2 \exp\!\Big(-\tfrac{\|x-x'\|^2}{2\ell^2}\Big) \;+\; \sigma_n^2\, \delta_{xx'}, \tag{1}$$

i.e. a constant-scaled RBF plus a white-noise term. Hyperparameters $\{\sigma_f, \ell, \sigma_n\}$ are obtained by maximising the log-marginal likelihood with two random restarts.

C. Gaussian Process regressor for time

For dating an unseen passage we regress the (continuous) year of authorship on the same topic vector. Because the year axis is short (2007–2024) and the underlying topic drift is expected to be smooth but not infinitely differentiable, we adopt a Matérn-2.5 kernel,

$$k_{\nu=5/2}(x, x') \;=\; \Big(1 + \tfrac{\sqrt{5}r}{\ell} + \tfrac{5r^2}{3\ell^2}\Big) \exp\!\Big(-\tfrac{\sqrt{5}r}{\ell}\Big), \tag{2}$$

with $r = \|x - x'\|$, again coupled with an explicit white-noise term.

D. Baselines and evaluation

For classification we compare against ℓ_2-regularised logistic regression; for regression we compare against ridge regression. All models are evaluated with five-fold cross-validation, stratified by class for the classifier and shuffled for the regressor. We report accuracy, macro-F_1, mean absolute error (MAE) and R^2 together with cross-fold standard deviations.

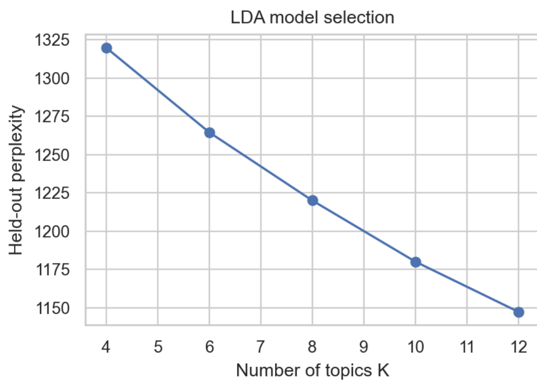

Fig. 1: Held-out perplexity as a function of the number of topics K. Perplexity falls monotonically but with diminishing returns; $K=8$ is chosen for interpretability.

V. Experimental Setup

The end-to-end pipeline is implemented in Python 3.12 with scikit-learn 1.8, numpy, pandas, matplotlib, seaborn, nltk, wordcloud and pypdf, managed through the uv package manager. All experiments were run on a macOS workstation with an Apple Silicon CPU; total wall-clock time, including downloads, is under four minutes. Random seeds are fixed at 42 throughout.

To select the number of topics K, we sweep $K \in \{4, 6, 8, 10, 12\}$ and record the held-out perplexity reported by the LDA implementation. As Fig. 1 shows, perplexity declines monotonically. We therefore choose $K = 8$ as a compromise between predictive accuracy and human interpretability; beyond eight topics, additional components either fragment existing themes or attach to publication boilerplate that survived the stopword pass.

VI. Results

A. Discovered topics

Table II lists the ten highest-probability terms for each of the eight topics, and Fig. 2 shows corresponding word-clouds. The themes are crisply recognisable to a reader familiar with Nepali planning discourse:

- T1 – Fiscal accounts. Revenue handles, grants, capital and recurrent loans – the bookkeeping vocabulary of the Ministry of Finance.
- T2 – Macroeconomy. Economic growth, investment, employment and rates – the language used to motivate sectoral reform.
- T3 – Trade, energy and provinces. Tourism, exports, industrial supply and provincial industry, reflecting the post-federal industrial agenda.
- T4 – Infrastructure. Roads, bridges, electricity transmission, drinking water; consistently the largest single capital-spending category.
- T5 – Human development. Health, education, women, children, citizens' rights – the social spending block.

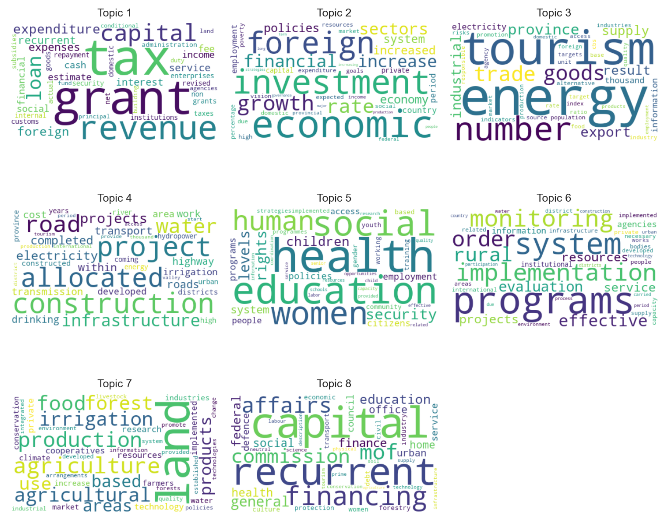

Fig. 2: Word-clouds for the eight LDA topics. Larger words have higher probability within the topic.

- T6 – Implementation and governance. Monitoring, evaluation, agencies, institutional reform.
- T7 – Agriculture and land. Irrigation, forests, cooperatives, food production.
- T8 – Federal financial structure. Recurrent and capital splits, Federal Affairs, the Office of the Auditor-General – a topic that, as we show below, barely existed before 2015.

B. Plans versus budget speeches

Fig. 3 contrasts the average topic probability for Plan chunks and Speech chunks. Plans are dominated by T2 (macroeconomy), T5 (human development), T6 (governance) and T7 (agriculture); Budget Speeches are dominated by T1 (fiscal accounts), T4 (infrastructure) and T8 (federal finance). This is precisely the division of labour one would expect: Plans articulate medium-term strategy in broad sectoral language, while Speeches articulate fiscal-year allocations in line-item language.

C. Gaussian Process classification

With these eight features the GP classifier separates the two genres at a cross-validated accuracy of 0.907 ± 0.028 and macro-F_1 of 0.901, compared with 0.820 ± 0.025 accuracy and 0.798 F_1 for logistic regression on the same features (Table III). Fig. 4 shows the confusion matrix on the full training set (210+132 correct, three errors). The misclassified chunks turn out to belong to the macroeconomic overview sections of budget speeches, which are written in the same strategic register as Plans.

The t-SNE projection in Fig. 5 provides geometric intuition for the result: in eight-topic space the two genres already separate into distinct clusters, leaving only a

TABLE II: Top eight words for each LDA topic ($K = 8$).

Topic	Top words
T1 (Fiscal)	tax, grant, revenue, capital, loan, expenditure, foreign, expenses
T2 (Macroeconomy)	economic, investment, foreign, growth, rate, financial, increase, sectors
T3 (Trade/Energy)	energy, tourism, goods, trade, province, export, industrial, supply
T4 (Infrastructure)	construction, project, road, water, infrastructure, electricity, completed, transport
T5 (Human dev.)	health, education, social, women, human, security, rights, children
T6 (Governance)	programs, system, implementation, monitoring, rural, evaluation, projects, service
T7 (Agriculture)	land, agriculture, production, food, agricultural, irrigation, forest, areas
T8 (Federal fin.)	capital, recurrent, financing, affairs, commission, federal, finance, education

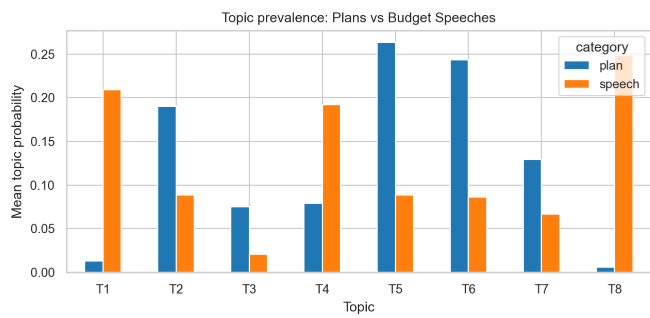

Fig. 3: Mean topic probability per category. Plans emphasise sectoral and governance topics; Budget Speeches emphasise fiscal, infrastructure and federal-finance topics.

TABLE III: Classification of Plans vs Budget Speeches (5-fold CV).

Model	Accuracy	Macro F_1
Logistic Regression	0.820 ± 0.025	0.798 ± 0.032
GP (RBF)	$\mathbf{0.907 \pm 0.028}$	$\mathbf{0.901 \pm 0.032}$

narrow contact region for the GP decision boundary to negotiate.

D. Dating a passage

Fig. 6 traces the mean topic probability for each year of the corpus. Two qualitative observations stand out. First, T8 (federal finance) is essentially absent before 2015 and rises sharply after the constitution-mandated federal transition. Second, T2 (macroeconomy) declines visibly after 2018 as the documents pivot from aspirational growth narratives to fiscal-recovery narratives in the COVID-19 and post-COVID years.

These shifts are large enough that a GP regressor can recover the year of a passage from its topic mixture with a cross-validated MAE of 3.48 ± 0.15 years and $R^2 = 0.353 \pm 0.138$, beating ridge regression (MAE $= 3.70$, $R^2 = 0.324$) by a small but consistent margin (Table IV). Fig. 7 plots predicted year against true year; the $\pm 1\sigma$ band widens around documents from the 2016–2017 transition years, exactly where the discourse is most heterogeneous.

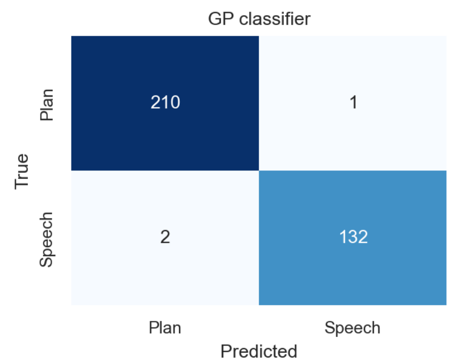

Fig. 4: Confusion matrix of the GP classifier on the full corpus.

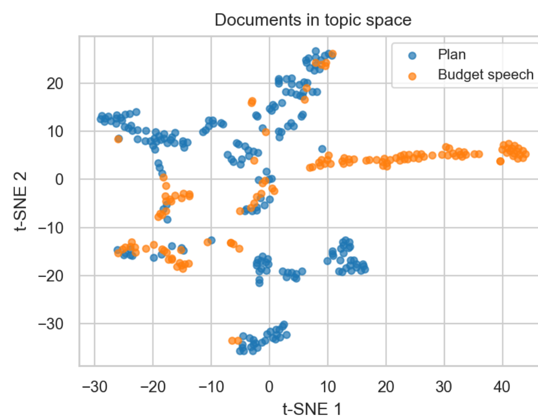

Fig. 5: Two-dimensional t-SNE embedding of the per-chunk topic distributions, coloured by document category.

VII. Social, Ethical, Legal and Professional Considerations

All texts analysed in this paper are official publications of the Government of Nepal and are released into the public domain for information and consultation; no copyright restrictions apply to research re-use, and we cite each download URL in the released sources file. Nevertheless, several considerations deserve explicit discussion.

Representation. Topic models are summaries, not summaries of intent. A topic labelled "human development"

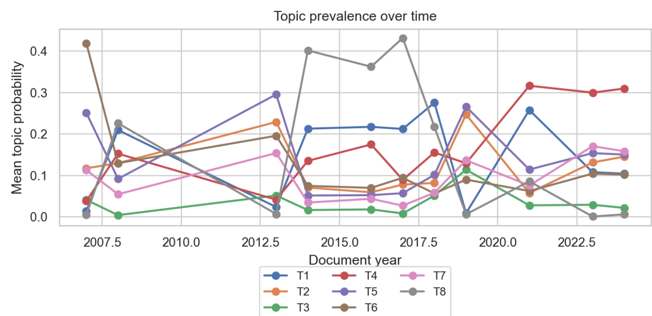

Fig. 6: Mean topic probability per document year, 2007–2024. Topic 8 (federal finance) emerges only after 2015; Topic 2 (macroeconomy) gradually loses share.

TABLE IV: Year regression on topic features (5-fold CV).

Model	MAE (years)	R^2
Ridge regression	3.70 ± 0.12	0.324 ± 0.102
GP (Matérn-5/2)	$\mathbf{3.48 \pm 0.15}$	$\mathbf{0.353 \pm 0.138}$

captures the language a document uses about that subject, not the adequacy of the allocations behind it. Any policy claim built on the discovered themes must therefore be triangulated with expenditure data.

Translation bias. We use the official English versions of each document, which are translations of the authoritative Nepali texts. Translation choices made by government translators can alter emphasis, especially around culturally specific terms. Future work should re-run the pipeline on the Nepali originals using a Devanagari-aware tokeniser.

Federalism and provincial voices. Our corpus is exclusively federal-government. Provincial and local-government budgets, which have grown sharply in importance since 2017, are not represented. Generalising the conclusions to "Nepali policy priorities" tout court would therefore be premature.

Professional responsibility. Because the GP model returns a calibrated posterior probability, deployments of such a tool for journalistic or oversight purposes should communicate uncertainty rather than headline-grabbing point predictions. We follow this principle throughout the figures.

VIII. Discussion and Conclusions

Combining LDA with a Gaussian Process classifier exposes a clean quantitative distinction between Nepal's medium-term Plans and its annual Budget Speeches: a small, well-regularised GP recovers the distinction at 90.7% accuracy from only eight latent features. A Matérn-kernel GP regressor further confirms that the corpus has a non-trivial temporal signature – the topics shift in measurable ways across the constitutional and federal transitions of 2015–2018.

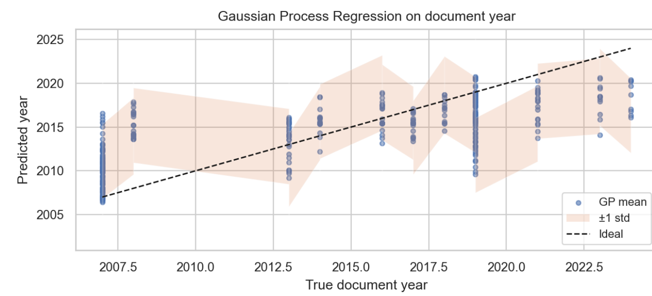

Fig. 7: Gaussian Process regression of document year on topic features. Shaded band is $\pm 1\sigma$ posterior uncertainty.

The GP outperforms the linear baselines on both tasks. The improvement is modest in absolute terms but matters substantively because the GP's posterior variance is itself an interpretable quantity: passages from years close to the federal transition come back with visibly wider posterior intervals, which is exactly the behaviour we want from an honest model in a regime change.

Limitations. Three plans and eight speeches are a thin slice of Nepal's planning history; extending the corpus to all sixteen Periodic Plans and to provincial budgets would strengthen every conclusion. Pre-processing in English loses lexical signal that the Nepali originals contain. Finally, LDA assumes exchangeable bags of words and therefore cannot capture argument structure.

Future work. We see two natural extensions. First, substituting a dynamic topic model [9] for static LDA would let topic-word distributions themselves evolve through time. Second, the GP classifier could be re-used as an auditing tool for new budget speeches: a draft that scores as an outlier under the model is, by construction, breaking with the recent fiscal rhetoric and would warrant a closer human read.

Author Contributions

The two authors contributed jointly to problem formulation, corpus curation, and writing. Specific contributions are as follows. Sandesh Tamang led the data-engineering work (PDF acquisition, text extraction, cleaning) and implemented the LDA component as the shared methodology. Aishwarya Rai led the probabilistic-modelling work, implementing both the Gaussian Process classifier on topic features and the Gaussian Process regressor for the temporal analysis as the individual technique, and produced the figures. Both authors jointly wrote the manuscript and reviewed the final results.

Reproducibility

The complete pipeline – download script, text extractor, analysis script, all figures, and the metrics file from which every number in this paper was drawn – is included in the appendix and in the project repository (policy_topic_modeling/). Random seeds are fixed at 42;

rerunning uv run python code/01_download.py followed by 02_extract.py and 03_analyze.py reproduces every result reported above.

References

[1] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," Journal of Machine Learning Research, vol. 3, pp. 993–1022, 2003.

[2] C. E. Rasmussen and C. K. I. Williams, Gaussian Processes for Machine Learning. MIT Press, 2006.

[3] K. M. Quinn, B. L. Monroe, M. Colaresi, M. H. Crespin, and D. R. Radev, "How to analyze political attention with minimal assumptions and costs," American Journal of Political Science, vol. 54, no. 1, pp. 209–228, 2010.

[4] J. Grimmer and B. M. Stewart, "Text as data: the promise and pitfalls of automatic content analysis methods for political texts," Political Analysis, vol. 21, no. 3, pp. 267–297, 2013.

[5] C. Lucas, R. A. Nielsen, M. E. Roberts, B. M. Stewart, A. Storer, and D. Tingley, "Computer-assisted text analysis for comparative politics," Political Analysis, vol. 23, no. 2, pp. 254–277, 2015.

[6] F. Moretti and D. Pestre, "Bankspeak: the language of World Bank reports, 1946–2012," New Left Review, vol. 92, pp. 75–99, 2015.

[7] C. K. I. Williams and D. Barber, "Bayesian classification with Gaussian processes," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 20, no. 12, pp. 1342–1351, 1998.

[8] M. L. Stein, Interpolation of Spatial Data: Some Theory for Kriging. Springer, 1999.

[9] D. M. Blei and J. D. Lafferty, "Dynamic topic models," in Proc. ICML, 2006, pp. 113–120.

[10] F. Pedregosa et al., "Scikit-learn: machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[11] National Planning Commission, Government of Nepal, The Fifteenth Plan, Fiscal Year 2019/20–2023/24. Kathmandu: NPC, 2020. [Online].

[12] Ministry of Finance, Government of Nepal, Budget Speech of Fiscal Year 2024/25. Kathmandu: MoF, 2024. [Online].